

BloodBridge - A Blood Donation Platform

1. Introduction

Access to timely and compatible blood donations can be the difference between life and death in emergency medical situations. Despite the presence of numerous donors across the country, the challenge lies in quickly identifying and connecting with the right donor. **BloodBridge** is an AI-powered platform designed to simplify and speed up the process of finding compatible blood donors. By combining blood group compatibility, geolocation, and blood rarity, it intelligently ranks and filters donors. The system uses a hybrid approach involving heuristics, geospatial calculations, and medical logic to break down the donor-matching task into smaller, solvable parts—making the process smarter, faster, and more reliable.

2. Problem Statement

The objective is to accurately match a blood recipient with the most suitable donor by integrating three core criteria: medical compatibility, geographic proximity, and donor prioritization based on blood group rarity. To approach this systematically, the problem is decomposed into three major computational components:

- **Medical Matching** using **Constraint Satisfaction Problem (CSP)**
- **Location-based Filtering** using the **Haversine Formula**
- **Donor Ranking** using a custom **Heuristic Score**

3. Implementation

The core functionality of the Bloodbridge platform was implemented using **Python** with **Streamlit** as the primary framework for the user interface.

3.1 User Interface and inputs

The application prompts the user to enter their name, age, required blood group, current city and state, and the maximum search radius. This input is collected via interactive Streamlit sidebar widgets and used to dynamically compute and display search results.

3.2 Data Integration

Two main CSV files are used:

- **Donor Dataset:** Contains donor details along with their city, blood group, contact info, hospital, and geolocation (latitude & longitude).

Name	Blood Group	State	City	Contact No.	Pincode	Email	Hospital Name	Latitude	Longitude
Hunar Dar	O-	maharashtra	mumbai	91388520713	23554	devanemir@koshy.com	Lilavati Hospital	19.0759837	72.8776559
Umang Kapoor	B+	karnataka	bengaluru	91676126981	448584	ihora@sandal.com	Fortis Hospital	12.9715987	77.5945627
Anay Tandon	A+	uttar pradesh	lucknow	91513571641	182968	kapurdrishya@ram.com	Medanta Lucknow	26.8466937	80.946166
Divya Sahai	O+	tamil nadu	chennai	91124626320	617612	anekdoughimmati@hotmail.com	Anella Chennai	13.0826603	80.2703164

- **City Coordinates Dataset:** Maps Indian cities to their latitude and longitude to help geolocate the recipient's position.

id	city	district	std-code	state	gst-state-code	iso_3166-2	population	rank	latitude	longitude
1	Port Blair	South Andaman	3192	Andaman and Nicobar Islands	AN	IN-AN	108058	423	11.6233774	92.7264828
2	Adoni	Kurnool	8512	Andhra Pradesh	AD	IN-AP	166537	261	15.6322227	77.2728368
3	Amaravati	Guntur	866	Andhra Pradesh	AD	IN-AP	103000	446	16.574517	80.359012
4	Anantapur	Anantapur	8554	Andhra Pradesh	AD	IN-AP	268503	168	14.6818877	77.6005911
5	Bhimavaram	West	8816	Andhra Pradesh	AD	IN-AP	142280	319	16.544709	81.520593

3.3 Blood Matching as a Constraint Satisfaction Problem (CSP)

Blood transfusion follows strict compatibility rules that can be modeled as a Constraint Satisfaction Problem. In this formulation, **the recipient's blood group** acts as the variable, while the **available donor groups** represent the domain. Constraints are defined by medically accepted blood type compatibilities—for instance, a recipient with A+ blood can safely receive from A+, A-, O+, and O- donors.

To implement this, we maintain a dictionary-based lookup structure where each recipient type maps to its compatible donor types. When a request is made, the system filters potential donors by referencing this lookup table, ensuring that only those with medically valid blood types are considered in the subsequent stages.

3.4 Geographic Filtering using Haversine Formula

After filtering for medical compatibility, the next layer of filtering is based on location. Given that donors and recipients may be located in different cities, a real-world distance computation is necessary to ensure logistical feasibility. We use the **Haversine formula**, which calculates the shortest curved distance between two geographic points using their latitudes and longitudes.

This allows us to:

- Exclude donors located beyond a certain user-defined radius (e.g., 150 km)
- Prioritize those located nearby, thus increasing the likelihood of timely blood delivery
- Handle edge cases involving donors on the outskirts of metropolitan regions

By applying this formula, we ensure the system remains effective in real-world emergency scenarios where time is of the essence

```
# Blood compatibility
def get_compatible_blood_groups(blood_group):
    compatibility = {
        "O-": ["O-"],
        "O+": ["O-", "O+"],
        "A-": ["O-", "A-"],
        "A+": ["O-", "O+", "A-", "A+"],
        "B-": ["O-", "B-"],
        "B+": ["O-", "O+", "B-", "B+"],
        "AB-": ["O-", "A-", "B-", "AB-"],
        "AB+": ["O-", "O+", "A-", "A+", "B-", "B+", "AB-", "AB+"]
    }
    return compatibility.get(blood_group.upper(), [])
```

```
# Haversine formula for distance # CSP PART
def haversine(lat1, lon1, lat2, lon2):
    R = 6371
    lat1, lon1, lat2, lon2 = map(radians, [lat1, lon1, lat2, lon2])
    dlat = lat2 - lat1
    dlon = lon2 - lon1
    a = sin(dlat/2)**2 + cos(lat1) * cos(lat2) * sin(dlon/2)**2
    c = 2 * asin(sqrt(a))
    return R * c
```

3.5 Heuristic-Based Donor Ranking

Once the medically eligible and geographically viable donors are identified, they are ranked using a **custom heuristic score**. This ranking is critical when multiple donors are available within the acceptable proximity. The scoring function is designed to prioritize two main aspects:

- **Proximity:** Donors closer to the recipient are favored for faster response
- **Blood Group Rarity:** Donors with rarer blood groups are given a higher weight to encourage rare donations

```
# Heuristic score with distance filter #HEURISTICS
def compute_scores(df, user_lat, user_lon, required_blood, max_distance_km=150):
    rare_weight = {
        "AB-": 1.5, "B-": 1.3, "A-": 1.3, "O-": 1.4,
        "AB+": 1.0, "A+": 0.8, "B+": 0.8, "O+": 0.7
    }

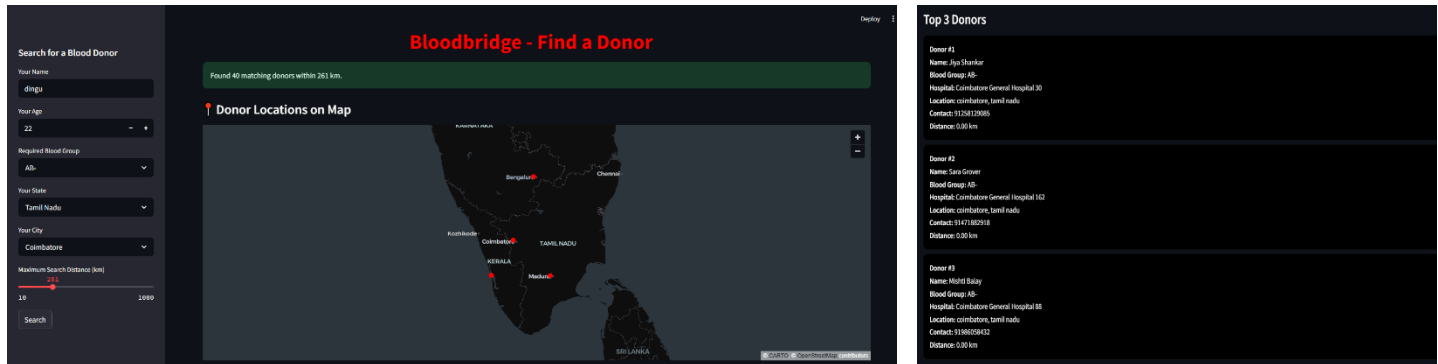
    scores = []
    distances = []
    for _, row in df.iterrows():
        d = haversine(user_lat, user_lon, row['Latitude'], row['Longitude'])
        distances.append(d)
        rarity_score = rare_weight.get(row['Blood Group'], 1.0)
        score = (1 / (d + 0.1)) + rarity_score
        scores.append(score)

    df = df.copy()
    df['Distance (km)'] = distances
    df['Heuristic Score'] = scores
    df = df[df['Distance (km)'] <= max_distance_km]
    df.sort_values(by='Heuristic Score', ascending=False, inplace=True)
    return df
```

3.6 Visualization and Results

The platform provides a user-friendly interface with multiple ways to view and interpret donor information:

- **Map View:** Donor locations are plotted on an interactive map using PyDeck, centered around the recipient's city. This allows users to visually assess the proximity of available donors in real time.
- **Ranked Donor Table:** A detailed table lists donors sorted by a heuristic score, which considers both blood type rarity and geographical distance. Key information such as name, blood group, hospital, contact details, and distance is displayed.
- **Top 3 Donor Cards:** The three most suitable donors are highlighted separately in visually distinct cards, providing a quick snapshot of their details for immediate contact. This makes the interface highly informative and user-friendly.



Ranked Donor List								
	Name	Blood Group	State	City	Contact No.	Email	Distance (km)	Hospital Name
0	Jiya Shankar	AB-	tamil nadu	coimbatore	91258129085	alisha68@yahoo.com	0	Coimbatore General Hospital 30
1	Sara Grover	AB-	tamil nadu	coimbatore	91471882918	jivin23@dixit-grover.com	0	Coimbatore General Hospital 162
2	Mishti Balay	AB-	tamil nadu	coimbatore	91986058432	priyansh16@hotmail.com	0	Coimbatore General Hospital 88
3	Shaan Mand	AB-	tamil nadu	coimbatore	91478178294	srana@mander-handa.com	0	Coimbatore General Hospital 50
4	Khushi Lanka	AB-	tamil nadu	coimbatore	91203294700	romilbrahmhatt@shetty.net	0	Coimbatore General Hospital 154
5	Darshit Chandran	O-	tamil nadu	coimbatore	91118234025	venkataraman@hotmail.com	0	Coimbatore General Hospital 42

4. Conclusion

BloodBridge demonstrates how AI principles can enhance healthcare accessibility. Through simple UI, geolocation, and intelligent scoring, it enables users to efficiently locate suitable blood donors. The combination of heuristic prioritization and compatibility filtering ensures relevance and urgency in donor selection. This project showcases a successful fusion of practical healthcare needs with AI-driven technology.

5. Future Scope

- **Live Donor Availability:** Integrating donor response status or last donation date to ensure up-to-date availability.
- **Real-Time Alerts:** Notify potential donors via SMS or app notifications.
- **Urgency-Based Scoring:** Prioritize donors based on the severity or urgency of a case.